



Machine Learning Lab Final Exam (Part 2)

Report

Hybrid Vision Transformer + Quantum Neural Network for Skin Disease
Classification

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Batch / Section	BSCS - CS 04 B
Course	Machine Learning
Dataset	Multi-Class Viral Skin Lesion Dataset (MCVSLD) — 6 Classes
Platform	Google Colab
Date	June 2026

1. Project Overview

The notebook implements a hybrid deep learning architecture combining a pretrained Vision Transformer (ViT-B/16) with a Quantum Neural Network layer. The model performs multi-class skin disease classification across six categories. Transfer learning is used by freezing the ViT backbone and training a lightweight quantum-classical classifier.

2. Training Configuration

- Pretrained ViT backbone (frozen)
- Weighted loss and weighted sampler for class imbalance
- AdamW optimizer
- Learning rate scheduling
- Early stopping

- Data augmentation for training images
- Quantum layer integrated after feature reduction

3. Training Performance

Training logs show steady improvement from approximately 35% training accuracy and 60% validation accuracy in the first epoch to around 86–89% validation accuracy before convergence.

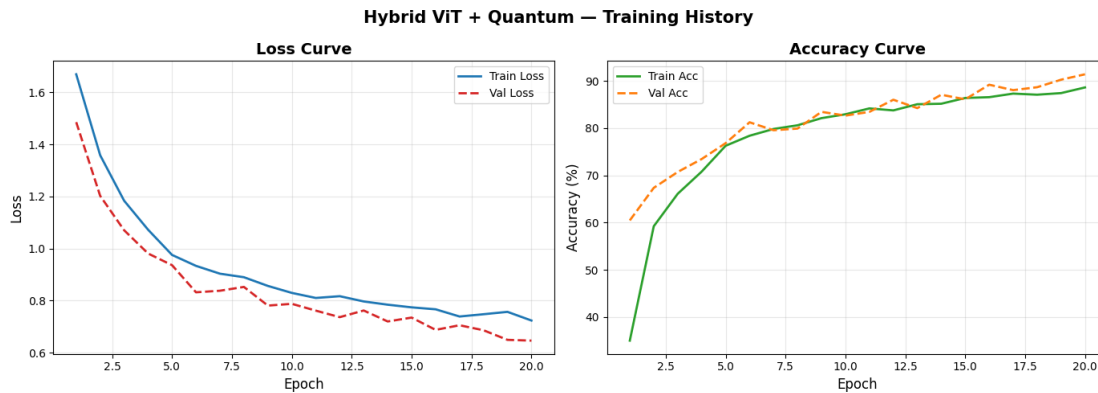


Figure 1: Training and validation learning curves.

4. Final Test Evaluation

Test Accuracy: 90.21%
 Macro Precision: 87.79%
 Macro Recall: 91.75%
 Macro F1-Score: 89.54%

The model generalizes effectively on unseen test data and achieves strong performance across all six disease categories.

5. Class-wise Results

Class	Precision	Recall	F1	Support
Chickenpox	0.79	0.88	0.83	113
Cowpox	0.81	0.98	0.89	99
HFMD	0.94	0.97	0.96	242
Healthy	0.90	0.92	0.91	171

Measles	0.87	0.92	0.89	83
Monkeypox	0.95	0.84	0.89	426

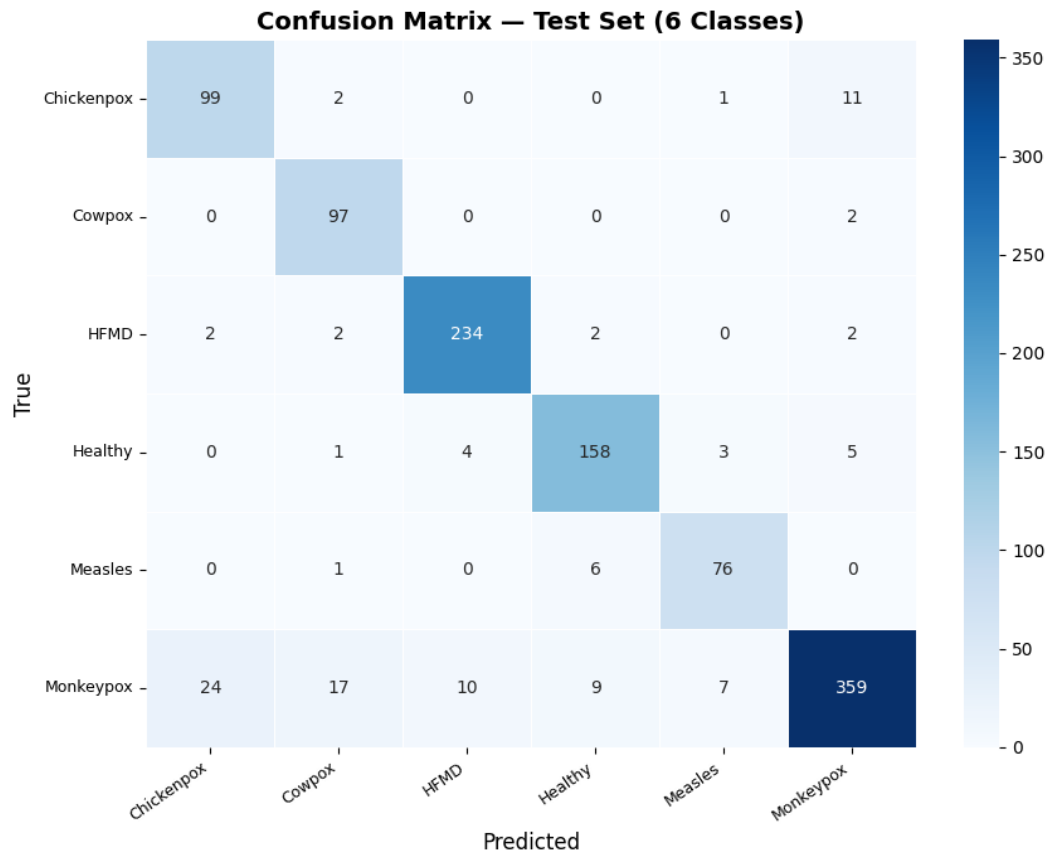


Figure 2: Confusion matrix illustrating prediction quality across disease categories.

6. Explainable AI (XAI) Results

Prediction visualizations demonstrate that the trained model correctly identifies multiple disease categories from unseen samples.



Figure 3: Sample predictions on test images.

CLS token activation analysis provides insight into feature representations learned by the Vision Transformer.

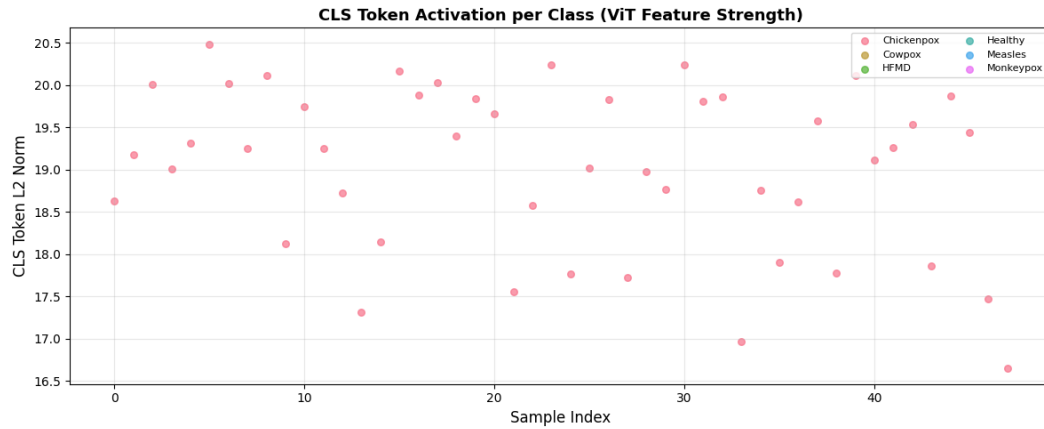


Figure 4: CLS token activation analysis.

Class-wise accuracy analysis shows balanced performance across disease categories.

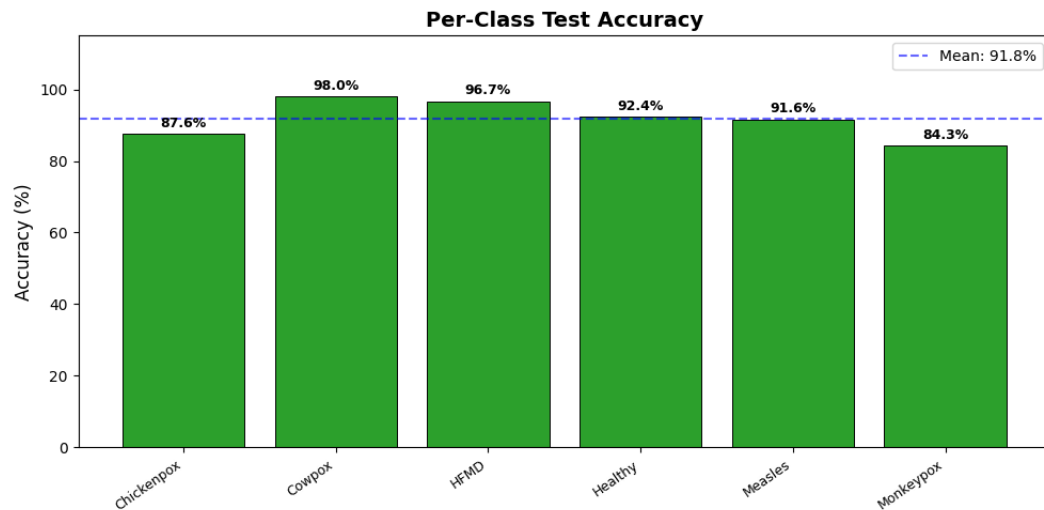


Figure 5: Class-wise accuracy comparison.

7. Strengths of the Proposed Model

- High overall accuracy (90.21%)
- Strong recall across all classes
- Effective handling of class imbalance
- Reduced trainable parameters due to frozen ViT backbone
- Explainable AI visualizations included
- Innovative hybrid quantum-classical architecture

8. Limitations

- Chickenpox and Cowpox classes show lower precision than other classes.
- Quantum layer increases computational complexity.
- Performance depends on image quality and dataset diversity.

9. Conclusion

The Hybrid ViT-QNN model successfully classifies six skin disease categories with 90.21% test accuracy and strong macro F1-score. The integration of transfer learning, quantum computing concepts, class balancing strategies, and explainable AI makes the system suitable as an advanced machine learning laboratory project.
